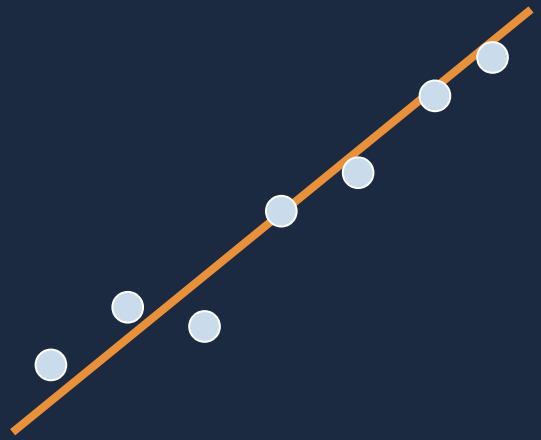
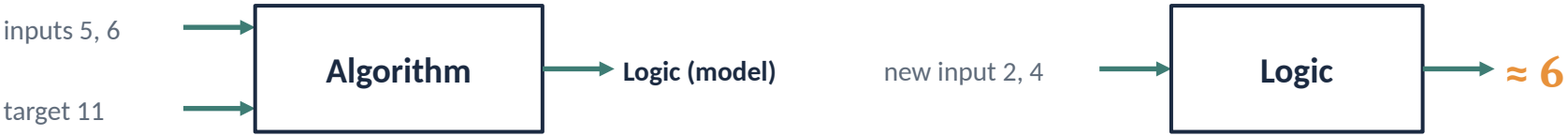
Linear Regression



What is supervised learning?

- Training data is labelled: every example has inputs and a target (label)
- Goal: learn a mapping function that predicts the target for new, unseen inputs
- Example — addition: inputs 5 and 6, target = 11
- Works even for complicated, high-dimensional input data



Training with training data

Predicting with new data

We may not get exactly 6 — a value close to 6, depending on the training data and algorithm.

Two types of supervised learning

Supervised Learning

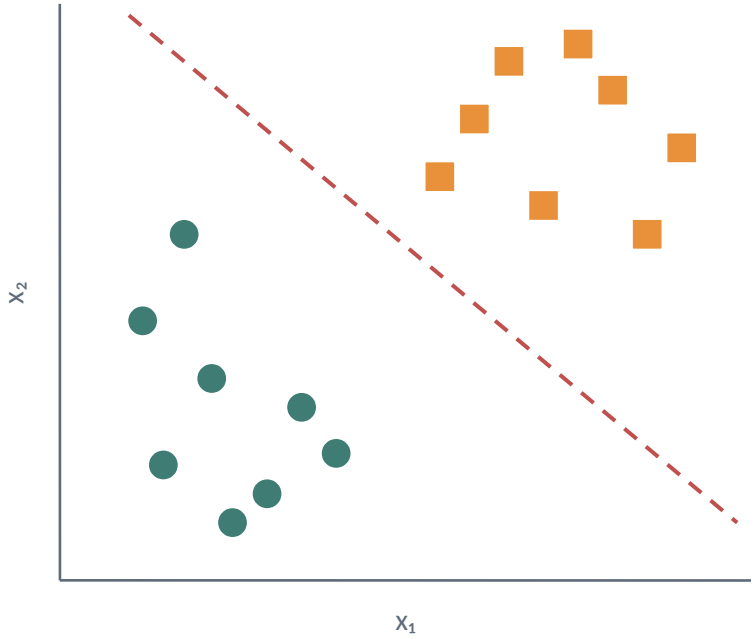
Regression

- Predicts a continuous value (−∞ to +∞)
- Price of a house in a city? Value of a stock?
- Total runs on board in a cricket game?
- Deals with the magnitude of the response

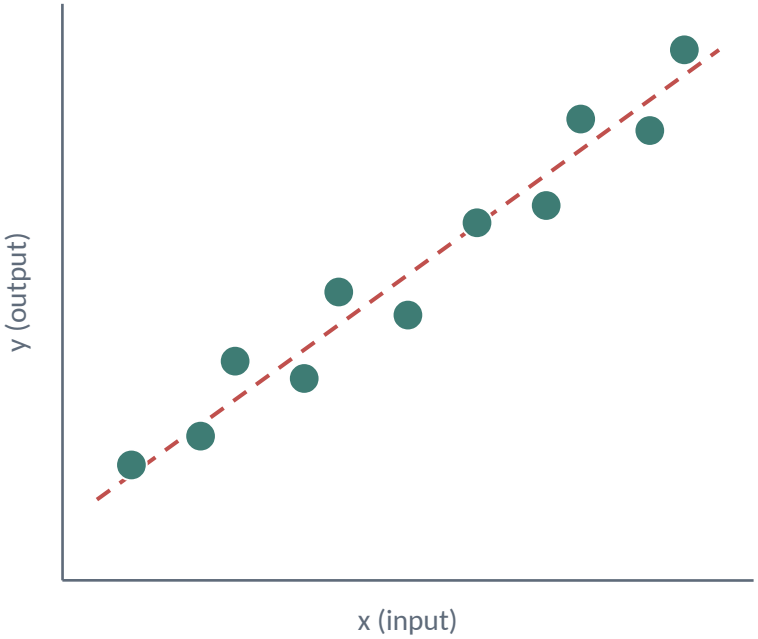
Classification

- Predicts a discrete category (class)
- Binary: spam or not? rain today or not? cat or not?
- Multi-class: spam / important / promotion? cat / dog / tiger?
- Assigns each input to a specific group

Classification separates; regression fits



Classification — find a boundary



Regression — fit a line through the points

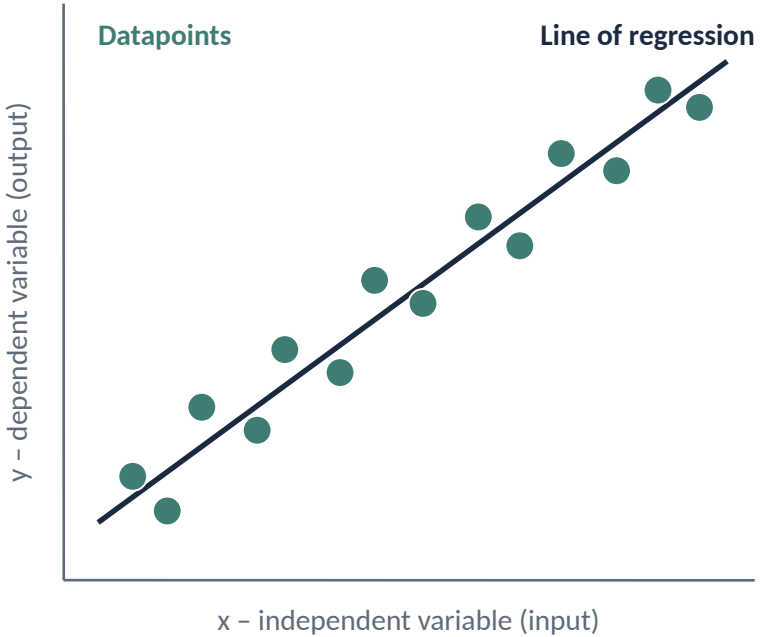
Linear Regression

The relationship between the dependent and independent variable(s) is linear — find the line (or hyperplane) of best fit

Linear regression: the idea

Fit the data with the best line / hyperplane that "goes through" the points

- Models a linear relationship between a dependent variable (y, output) and one or more independent variables (x, inputs)
- Output is continuous → a form of function approximation
- Many lines are possible — we need the one with minimum total error: the line of regression
- House price dataset: Area, Bedrooms, Age, Location = independent variables; Price = dependent variable



Simple linear regression equation

One independent variable x

$$Y_i = \beta_0 + \beta_1 X_i$$

Y_i Dependent variable	β_0 Constant / intercept	β_1 Slope / coefficient	X_i Independent variable

The two parameters we learn (used throughout this lecture):

Parameter	Name	Meaning
β_0	Intercept	Predicted value $\hat{y}$ when x = 0 (where the line crosses the y-axis)
β_1	Slope	Change in $\hat{y}$ for a one-unit increase in x
$\hat{y} = \beta_0 + \beta_1 x$	Prediction	Value on the fitted line for a given x

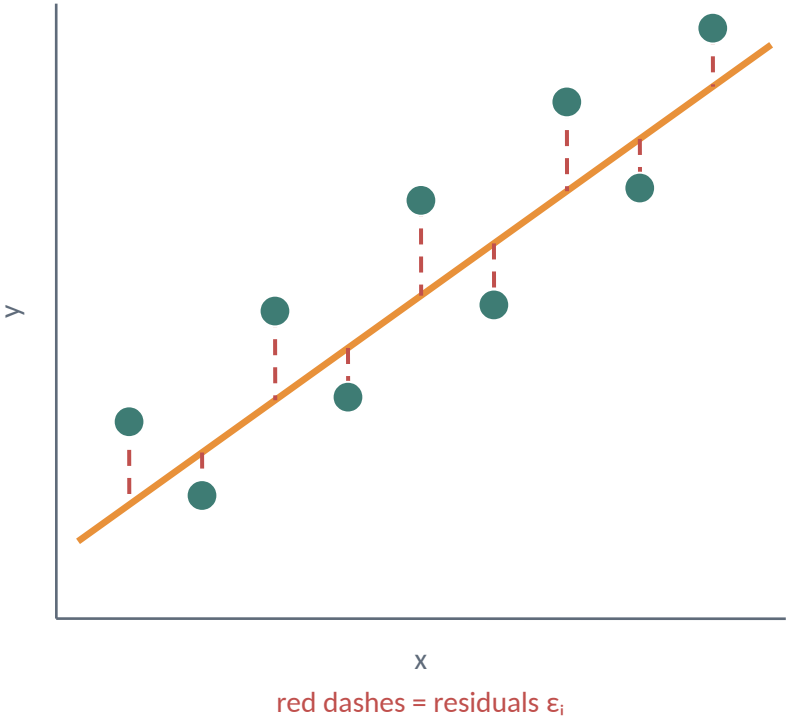
Residual (error) of each point

Every datapoint is approximated by the line; the gap is the residual

$$\varepsilon_i = y_i - \hat{y}_i$$

Symbol	Meaning
ε_i	Error term for a single predicted value
y_i	Actual (observed) value
$\hat{y}_i$	Estimated (predicted) value on the line

Actual = Predicted + Residual
Residuals can be negative, so we square them before adding; the best β_0, β_1 make the total squared error minimum.



red dashes = residuals ε_i

Cost function $J(\beta_0, \beta_1)$

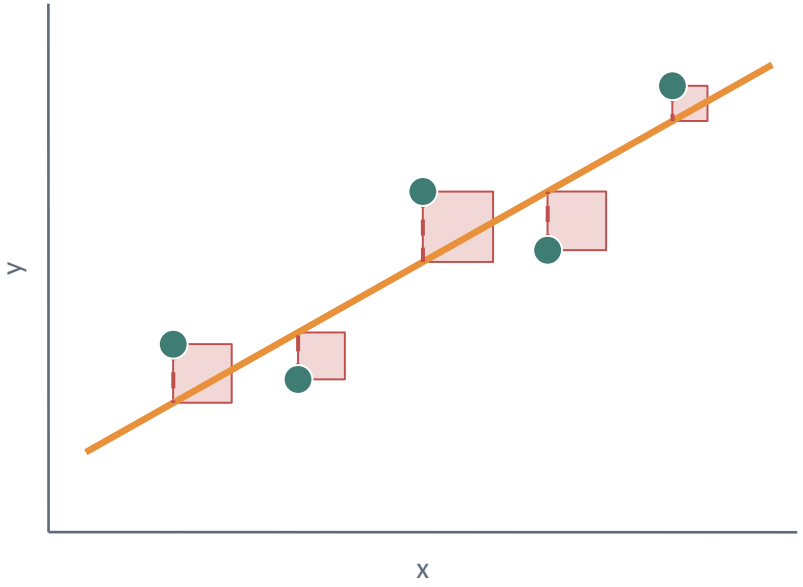
One number that scores how well a line fits all the data

$$J(\beta_0, \beta_1) = \sum_{i=1}^{n} (y_i - \hat{y}_i)^2$$

n	y_i	$\hat{y}_i = \beta_0 + \beta_1 x_i$
number of datapoints	actual value	predicted value

Goal: choose the β_0, β_1 that make J as small as possible.

- Square each error (y − ŷ), then add them up for i = 1 to n.
- Squaring makes every error positive and punishes large errors more.
- The line with the lowest cost is the line of regression.



Each red square has area $(y_i - \hat{y}_i)^2$; J adds up all the squares.

Regression Metrics

Evaluating the accuracy of the model

Metric	Good model
SSE (Sum of Squared Errors)	Low (close to 0)
MSE (Mean Squared Error)	Low (close to 0)
RMSE (Root Mean Squared Error)	Low (close to 0)
MAE (Mean Absolute Error)	Low (close to 0)
MAPE (Mean Absolute Percentage Error)	Low (below 10% is very good)
R^2 (Coefficient of Determination)	High (close to 1)

Classification Metrics

Evaluating the accuracy of the model

Metric	Good model
Accuracy	High (close to 1 or 100%)
Precision	High (close to 1)
Recall (Sensitivity)	High (close to 1)
Specificity	High (close to 1)
F1-score	High (close to 1)

Least-squares method: finding β_0 and β_1

For the model $\hat{y} = \beta_0 + \beta_1 x$

1. **Means**

$$\bar{x} = (1/n)\,\Sigma\, x_i \qquad \bar{y} = (1/n)\,\Sigma\, y_i$$

2. **Variance of x**

$$\mathrm{Var}(x) = 1/(n-1)\cdot\Sigma\,(x_i - \bar{x})^2$$

3. **Covariance of x and y**

$$\mathrm{Cov}(x, y) = 1/(n-1)\cdot\Sigma\,(x_i - \bar{x})(y_i - \bar{y})$$

4. **Slope**

$$\beta_1 = \mathrm{Cov}(x, y)\ /\ \mathrm{Var}(x)$$

5. **Intercept**

$$\beta_0 = \bar{y} - \beta_1\,\bar{x}$$

Worked example — Steps 1 to 3

Fit $\hat{y} = \beta_0 + \beta_1 x$ to five points: means, variance and covariance

i	x	y	x − x̄	y − ȳ	(x−x̄)(y−ȳ)	(x−x̄)²
1	1	1.00	−2	−1.06	2.12	4
2	2	2.00	−1	−0.06	0.06	1
3	3	1.30	0	−0.76	0.00	0
4	4	3.75	1	1.69	1.69	1
5	5	2.25	2	0.19	0.38	4
Σ	15	10.30			4.25	10

Step 1 — Means (n = 5)

$$\bar{x} = 15/5 = 3.0 \qquad \bar{y} = 10.30/5 = 2.06$$

Step 2 — Variance of x

$$\mathrm{Var}(x) = 10 \;/\; 4 = 2.5$$

Step 3 — Covariance

$$\mathrm{Cov}(x, y) = 4.25 \;/\; 4 = 1.0625$$

Divide by $n - 1 = 4$ for the sample variance and covariance.

Worked example — Steps 4 and 5

Slope β_1, intercept β_0 and the final model

Step 4 — Slope

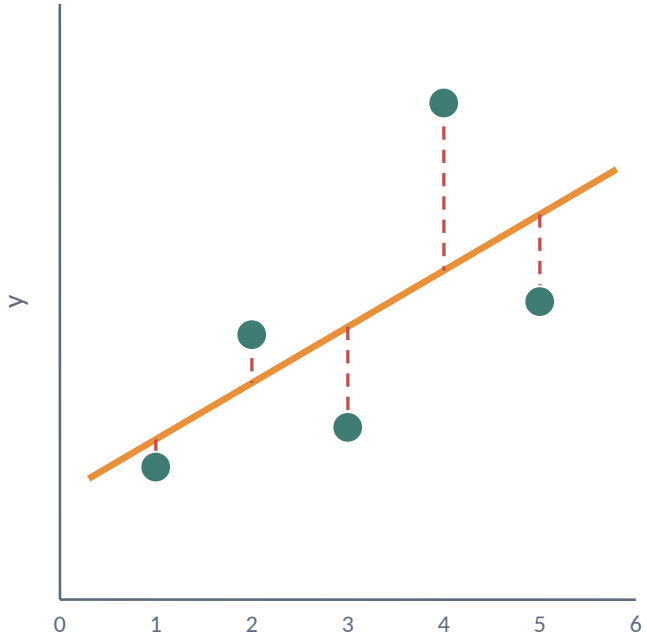
$$\beta_1 = \mathrm{Cov}(x, y) \,/\, \mathrm{Var}(x) = 1.0625 \,/\, 2.5 = 0.425$$

Step 5 — Intercept

$$\beta_0 = \bar{y} - \beta_1 \bar{x} = 2.06 - 0.425 \times 3.0 = 0.785$$

$$\hat{y} = 0.785 + 0.425x$$

Prediction for x = 6: $\hat{y} = 0.785 + 0.425 \times 6 = 3.335$



Multiple linear regression

More than one independent variable

Definition

When the dependent variable y depends linearly on two or more independent variables x_1, x_2, …, x_p, the model is called multiple linear regression. With p features we find β_0, β_1, …, β_p that minimise the squared error; the fitted surface is a hyperplane instead of a line.

$$\hat{y} = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \ldots + \beta_p x_p$$

$\hat{y}$ Predicted output	β_0 Intercept	$\beta_1 \ldots \beta_p$ Coefficient of each feature	$x_1 \ldots x_p$ Independent variables

Assumptions behind linear regression

Linearity

The relationship between a feature and the mean of the dependent variable is linear

y changes with x at a steady rate, so a straight line is the right shape for the data.

Independence

Each random sample (observation) is independent of the others

each observation is separate, and one data point doesn't influence another

Homoscedasticity (Equal variance)

Variance of the residuals is the same for any value of x

The spread of the residuals (errors) is about the same for small and large x values.

Normality

For any fixed value of x, y is normally distributed

for any given x, the y values (equivalently, the residuals) follow a **normal, bell-shaped distribution** around the line. Most points are close to the line and few are far away.

Drawback: if any assumption is violated, the model's results are unreliable — and it is hard for all of them to hold true in real-world data.

When to use linear regression

Continuous outcome; relationship roughly a straight line

Predicting quantity

Forecasting sales, calls to a call centre, electricity consumption

Relationship analysis

Strength and direction of the effect of predictors on the outcome

Homoscedastic data

Residuals roughly equal across all levels of the inputs

Parametric assumptions met

Linearity, independence and normality are satisfied

Use cases

- House price prediction
- Identifying relationships
- Market trends and analysis
- Business risk assessment
- Scientific research
- Price estimation
- Understanding impact